A variance proof of Davis's polynomial-ratio bound

Principia Math In Progress

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MathDB 319187; Davis, Conjecture 3.1 [1].

Task 1. Prove the original bound

Write $(x)_{\underline{i}} = x(x-1)\cdots(x-i+1)$, with $(x)_{\underline{0}} = 1$, and let

$$P_d(x) = \sum_{i=0}^{d} \frac{(x)_{\underline{i}}^2 (x)_{\underline{d-i}}^2}{i!(d-i)!}, \qquad q_d(x) = \frac{2x^2 - 2dx}{d+1} + \frac{d}{2}.$$

The remainder in the source is $R_d = P_{d+1} - q_d P_d$: Lemma 1.1 of its journal version proves that q_d is the polynomial division quotient.

Theorem 1. *For every integer $d \geq 5$ and every real $x \geq (d + \sqrt{d+2})/2$,*

$$-\frac{R_d(x)}{P_d(x)} > \frac{199(d-2)}{200(2x+d-2)}. \tag{1}$$

Thus the conjectured weak inequality holds, with its exact constant $0.995 = 199/200$.

The proof is analytic for all $d \geq 40$. For the remaining 35 degrees, we give exact integer polynomial certificates valid on whole real rays, not pointwise numerical sampling.

The ratio as a variance

Put $c = x - d/2$ and introduce the nonnegative weights

$$w_i = \frac{(x)_{\underline{i}}^2 (x)_{\underline{d-i}}^2}{i!(d-i)!}, \qquad \Pr(I = i) = \frac{w_i}{P_d(x)} \quad (0 \leq i \leq d).$$

The denominator is positive: the middle summand has no vanishing factor when $x > d/2$. Since $w_i = w_{d-i}$, the variable $Z = I - d/2$ has mean zero. Write $V = \mathbb{E}Z^2$ and $M_4 = \mathbb{E}Z^4$.

Adding a factor to either falling factorial gives

$$(d+1)P_{d+1}(x) = \sum_{i=0}^{d} w_i\big((x-i)^2 + (x-d+i)^2\big).$$

For example, the first sum contributes j times the jth summand of P_{d+1}, and the second contributes $d+1-j$ times that summand. Consequently

$$\frac{P_{d+1}}{P_d} = \frac{2c^2 + 2V}{d+1}, \qquad -\frac{R_d}{P_d} = \frac{d - 4V}{2(d+1)}. \tag{2}$$

There is also an exact fourth-moment identity. Define

$$a_i = (d-i)(x-i)^2, \qquad b_i = i(x-d+i)^2.$$

Direct cancellation in the weights proves $w_i a_i = w_{i+1} b_{i+1}$ for $0 \le i < d$; this is a polynomial identity and remains valid when some weights vanish. Since $a_d = b_0 = 0$, telescoping gives

$$\mathbb{E}\{a_I(f(I+1) - f(I)) + b_I(f(I-1) - f(I))\} = 0.$$

Use $f(i) = (i - d/2)^2$. In centered coordinates,

$$a_i - b_i = -2Z(c^2 + dc + Z^2), \qquad a_i + b_i = dc^2 + (d+4c)Z^2.$$

It follows that

$$4M_4 + AV = dc^2, \qquad A = 4c^2 + 4(d-1)c - d. \tag{3}$$

Cauchy–Schwarz gives $M_4 \ge V^2$. Thus, with $t = 4V$,

$$t^2 + At \le 4dc^2. \tag{4}$$

Every degree at least 40

Suppose $d \ge 40$. The required domain implies $c \ge \sqrt{d+2}/2 > 3$, so $A > 0$. Set

$$T = d - \frac{199(d+1)(d-2)}{200(c+d-1)}.$$

Here $T > d - (d+1)(d-2)/(d+2) = 3 - 4/(d+2) > 0$. We will prove $T^2 + AT > 4dc^2$. Since $y \mapsto y^2 + Ay$ is strictly increasing for $y \ge 0$, (4) will then imply $t < T$.

Put $u = d - 40 \ge 0$ and $v = c - 3 \ge 0$. An exact expansion is

$$40000(c+d-1)^2(T^2 + AT - 4dc^2) = C_0(u) + C_1(u)v + C_2(u)v^2 + C_3(u)v^3, \tag{5}$$

where

$$\begin{aligned}
C_0(u) &= 2201u^4 + 240358u^3 + 8708957u^2 + 110582964u + 241318564,\\
C_1(u) &= 800u^4 + 95400u^3 + 4131000u^2 + 80610800u + 677526400,\\
C_2(u) &= 1600u^3 + 196000u^2 + 8631200u + 136276800,\\
C_3(u) &= 800u^2 + 63200u + 1566400.
\end{aligned}$$

Every displayed coefficient is positive. Hence (5) is strictly positive, proving $4V = t < T$. By (2),

$$-\frac{R_d}{P_d} > \frac{199(d-2)}{400(c+d-1)} = \frac{199(d-2)}{200(2x+d-2)}.$$

In fact this part of the proof works whenever $d \ge 40$ and $c \ge 3$.

Exact certificates for degrees 5 through 39

Fix such a degree and put

$$b = \lfloor d/2 \rfloor, \quad q = \lceil d/2 \rceil, \quad K = 1000d + \left\lfloor \sqrt{10^6(d+2)} \right\rfloor, \quad X = 2000x - K.$$

The source domain implies $X \geq 0$. Define the degree-b integer polynomial

$$A_i(X) = \prod_{j=0}^{\min(i,d-i)-1} (X + K - 2000j) \prod_{j=q}^{\max(i,d-i)-1} (X + K - 2000j),$$

where an empty product is one, and set

$$S(X) = \sum_{i=0}^{d} \binom{d}{i} A_i(X)^2, \qquad N(X) = \sum_{i=0}^{d} \bigl(d - (2i-d)^2\bigr) \binom{d}{i} A_i(X)^2.$$

Factoring the common square $(x)_{\underline{q}}^2$ from the weights shows that $S > 0$ and $N/S = d - 4V$. Thus (1) is equivalent to positivity of

$$E_d(X) = (X + K + 1000(d-2))N(X) - 1990(d+1)(d-2)S(X). \tag{6}$$

For each of the 35 indicated degrees, every coefficient of E_d is nonnegative and its constant coefficient is positive. The degree is $2\lfloor d/2 \rfloor$; there are 787 coefficients in total. These finite integer assertions are verified by the complete checker below. The coefficient vectors, divided by their positive integer contents, are also recorded in `polynomial-certificates.json`.

```
from math import comb, isqrt

def mul(p, q):
    r = [0] * (len(p) + len(q) - 1)
    for i, a in enumerate(p):
        for j, b in enumerate(q):
            r[i+j] += a*b
    return r

total = 0
for d in range(5, 40):
    b, q = d//2, (d+1)//2
    K = 1000*d + isqrt(1000000*(d+2))
    S, N = [0]*(2*b+1), [0]*(2*b+1)
    for i in range(d+1):
        p = [1]
        J = list(range(min(i,d-i))) + list(range(q,max(i,d-i)))
        for j in J:
            p = mul(p, [K-2000*j, 1])
        for j, a in enumerate(mul(p,p)):
            w = comb(d,i)*a
            S[j] += w
            N[j] += (d-(2*i-d)**2)*w
    E = mul(N, [K+1000*(d-2), 1])
    for j, a in enumerate(S):
        E[j] -= 1990*(d+1)*(d-2)*a
    while len(E)>1 and E[-1] == 0:
        E.pop()
    assert len(E)-1 == 2*b
    assert E[0] > 0 and all(a >= 0 for a in E)
    total += len(E)
assert total == 787
```

This code uses only exact integer arithmetic from Python's standard library. Its coefficient assertions imply $E_d(X) > 0$ for every real $X \geq 0$, and therefore prove (1) for every real argument in the required domain. Together with the large-degree argument, this completes the proof. □

Scope and verification. The proof handles every integer degree $d \geq 5$, includes the stated cutoff, and proves a strict inequality. The accompanying verifier checks the universal symbolic identities, the 17 positive terms in (5), all 35 small-degree certificates, and 70 exact rational evaluations from the original falling-factorial definition. Those last evaluations check normalization only; the polynomial certificates prove the full real intervals. The same agent authored and audited the argument; independent peer review is not claimed.

Task 2. Determine the optimal universal constant

Determine the infimum, over the full source domain, of

$$\frac{2x + d - 2}{d - 2}\left(-\frac{R_d(x)}{P_d(x)}\right).$$

In particular, prove or refute the candidate that it is attained at $d = 27$ and $x = (27 + \sqrt{29})/2$. Obtain an exact algebraic description and a certified interval for the optimal constant.

Task 3. Remove the finite coefficient check

Use additional detailed-balance moment identities or a sharper lower bound for $\mathbb{E}Z^4$ to prove the stated 199/200 inequality uniformly from $d = 5$ onward. Seek a short analytic certificate that replaces the 35 polynomial expansions while preserving the full real-argument domain.

Task 4. Prove a uniform two-sided approximation

Derive explicit upper and lower error bounds for $-R_d/P_d - (d-2)/(2x+d-2)$, uniform for $x \geq (d + \sqrt{d+2})/2$. Determine the leading correction both when $x - d/2$ is proportional to $\sqrt{d}$ and when x/d stays above $1/2$, with rigorous control through the transition between them.

Task 5. Resolve the extremum conjecture

For odd $d \geq 5$, determine the number of real critical points of

$$Q_d(x) = 1 + \frac{R_d(x)}{P_d(x)}\frac{2x + d - 2}{d - 2},$$

interpreted after cancelling common factors. Prove or refute the source's single-maximum, single-minimum assertion and derive rigorous large-d expansions for their locations. This is Conjecture 3.2 in the journal version, numbered 3.9 in the preprint.

References

[1] D. M. Davis, *Maximizing a combinatorial expression arising from crowd estimation*, International Journal of Modern Mathematics **5** (2010), 129–148. Author-hosted journal version, revised 14 May 2010: `fredijmm.pdf`, Lemma 1.1 on PDF p. 5 and Conjecture 3.1 on PDF p. 12. Preprint: arXiv:0910.5725v1.